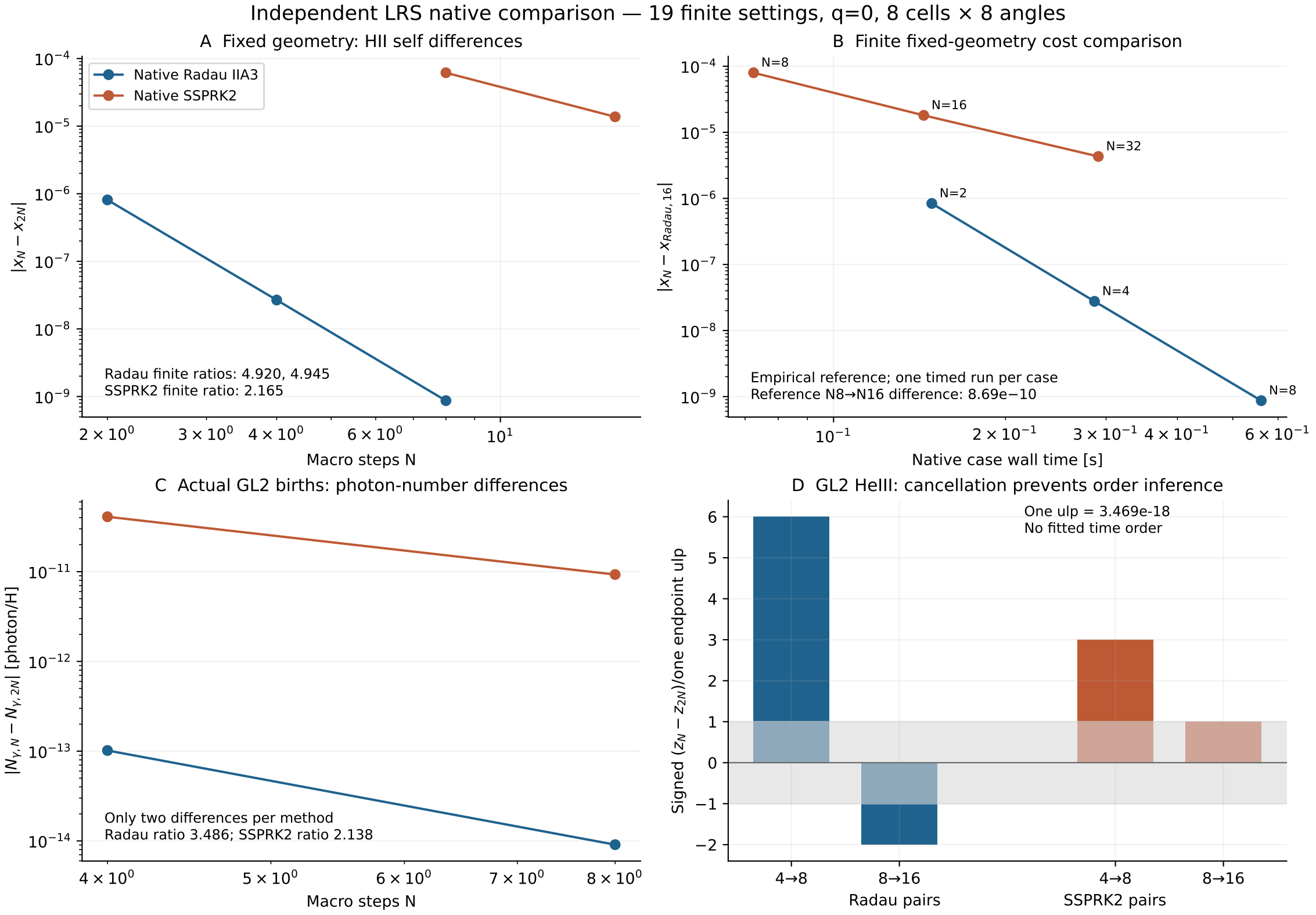
Independent LRS native comparison — 19 finite settings, q=0, 8 cells × 8 angles
A Fixed geometry: HII self differences
Native Radau IIA3
Native SSPRK2
$|x_N - x_{2N}|$
Radau finite ratios: 4.920, 4.945
SSPRK2 finite ratio: 2.165
Macro steps N
B Finite fixed-geometry cost comparison
N=8
N=16
N=32
N=2
N=4
N=8
$|x_N - x_{Radau,16}|$
Empirical reference; one timed run per case
Reference N8→N16 difference: 8.69e−10
Native case wall time [s]
C Actual GL2 births: photon-number differences
$|N_{\gamma,N} - N_{\gamma,2N}|$ [photon/H]
Only two differences per method
Radau ratio 3.486; SSPRK2 ratio 2.138
Macro steps N
D GL2 HeIII: cancellation prevents order inference
One ulp = 3.469e-18
No fitted time order
Signed $(z_N - z_{2N})$/one endpoint ulp
4→8
8→16
Radau pairs
4→8
8→16
SSPRK2 pairs